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1. What I have done

- Installed Blender, Godot with .NET support, Rider, and Insomnia.
- Set up the vitrivr engine first using Docker and later directly on my machine. During this process, I updated several configuration files to match the newer format.
- Tested Insomnia with POST requests and successfully received "retrievables" from the API.
- Started working through the Official Godot Tutorial. After completing the chapter "Jumping and squashing monsters", the project no longer started correctly. Therefore, I repeated the tutorial using the example project provided on GitHub
- Integration and Debugging of XR/OpenXR on Meta Quest 3
- Prototype progress:
  - Extracted the core ideas of the XRTools virtual keyboard and simplified its implementation to make appearance customization easier.
  - Created a scene where the user can type a word or text. After pressing Enter, a request is sent to a Flask Python server at: `http://<serverIP>:5050/search_one`
    The Flask server then forwards the query to the vitrivr engine:
    `http://<serverIP>:7070/api/sandbox/query`
    The Flask server returns a JSON response containing a URL pointing to the image:
    `http://<serverIP>:5050/media/<image>.jpg`
    The application then makes a second request to retrieve the image itself. The image is then displayed on a 2D mesh inside a 3D environment.
  - The frame automatically adjusts its size to the image dimensions. The scaling is currently not correct, so the frame thickness can differ between the horizontal and vertical sides.
  - All objects (except the ground) can currently be picked up. The keyboard and output image are hidden depending on user input.
  - Created a second scene containing a simple menu where the Flask server endpoint can be changed(Platform-Switching with double trigger, disabling player movements in menu).
- Provided my public SSH key and a sketch of the current architecture. The architecture is "work in progress" because I have not yet been able to test nginx (see last point in the next section).

2. Where I need support

- Access to BlenderKit and Polyhaven (paid products for now?).
- Streaming VR content to the Meta Quest from my MacBook is currently not possible because Meta Horizon Link is not available on macOS.
- Access to a Vive Focus 3/XR Elite headset
- Can I connect a Meta Quest or other VR devices to eduroam?
- Regarding vitrivr:
  At the moment, the vitrivr engine does not provide the real image path or filename for a retrievable. The VR application therefore only could receive the retrievable ID, but not the actual image (location). Because of this, I have not yet tested nginx because in my understanding nginx can only expose files that already exist in accessible directories via URLs. So I still need the Python/Flask server to map retrievable IDs to the actual image files? Is there a better way to obtain the original image path or filename from a retrievable?

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3. What I am planning to do for the next meeting

- ☒ Fix the frame scaling so that all borders have equal thickness
- ☒ Deploy prototype on the Vive devices
- ☒ Try running vitrivr on the server
- ☐ Cleanup Code
- ☐ Create a simple museum environment (room, lighting, etc.)
- ☐ Prepare for the next weekly meeting
- ☒ (Reduce the image retrieval process to a single request)
- ☒ (Test nginx integration)